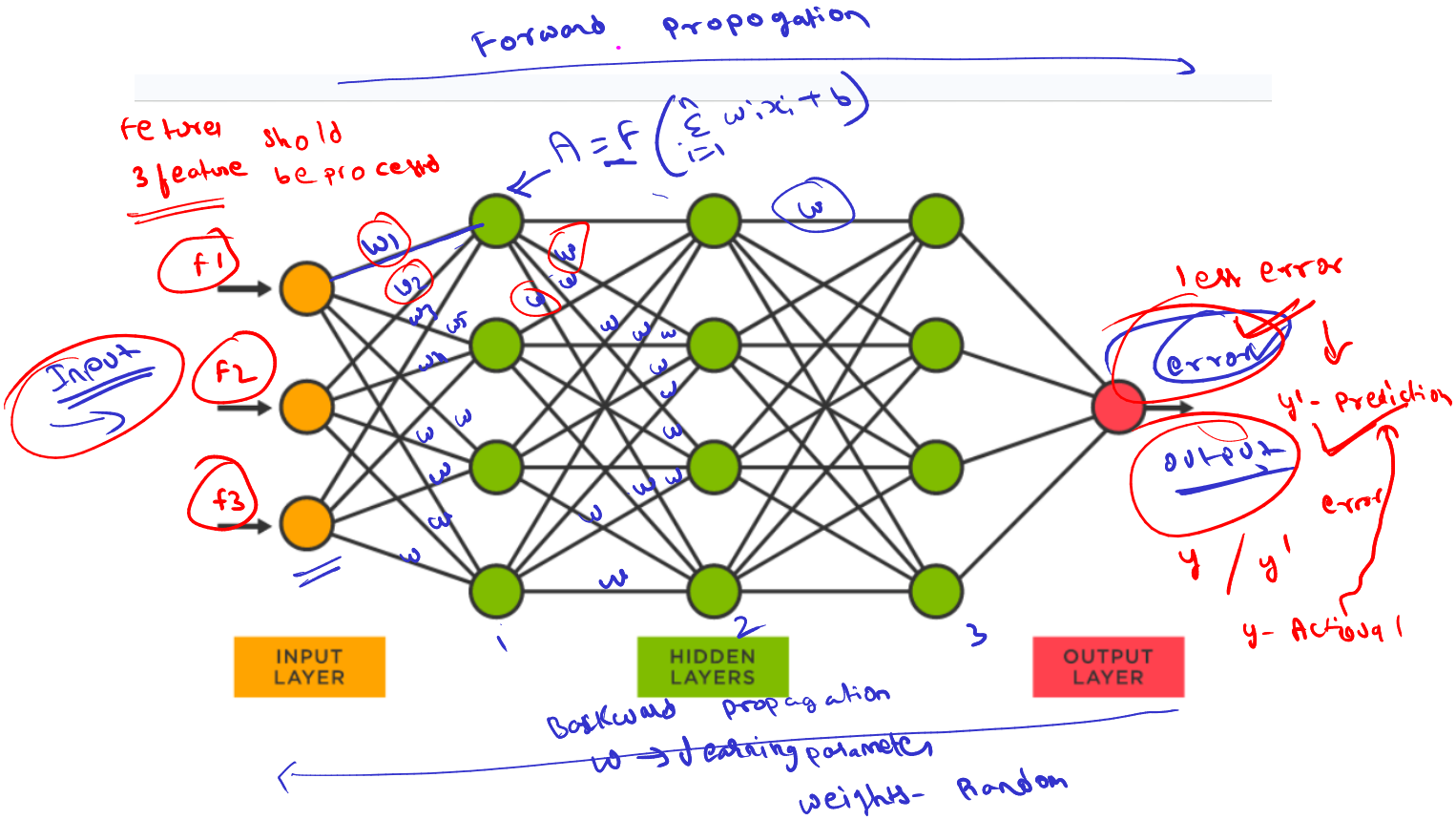
Neural Network

→ complex patterns
→ large dataset

Layers — each layer neurons / nodes

Input → no of features
hidden → Hyperparameter
output → type of Data



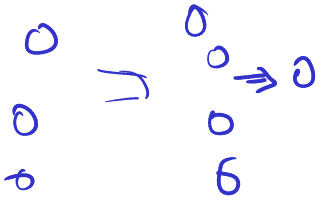
error → update the weights

CNN Convolution Neural Network

→ Images



⇒ Neural Net





→ it is difficult to extract features

Image → pixels → values

Feature → Neural Network

each pixel is a feature

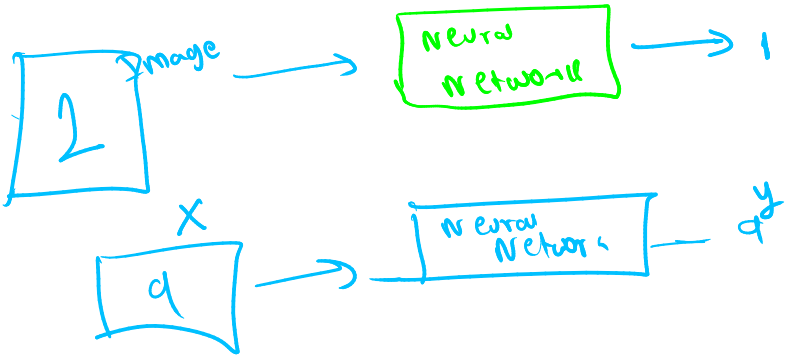
100 pixel →

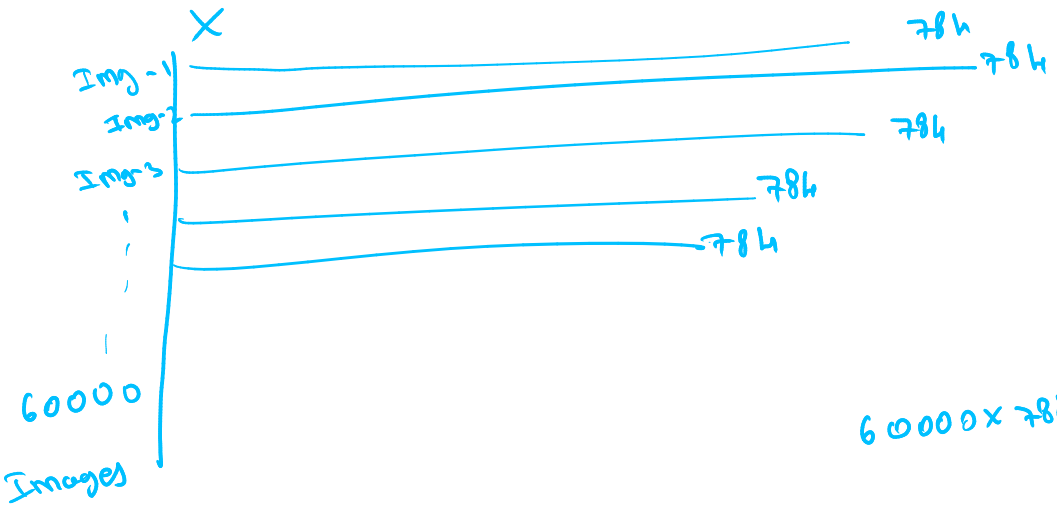
→ 0
0
0
0
0

100 Neuron

[ ] X → Images — pixels — 28×28
60000

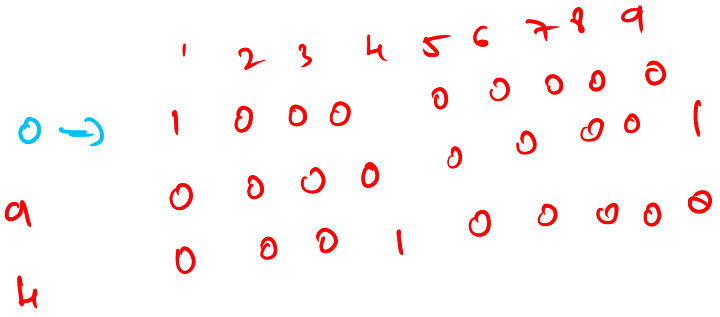
[ ] y → 0, 1, 2, 3, 4 - - 9

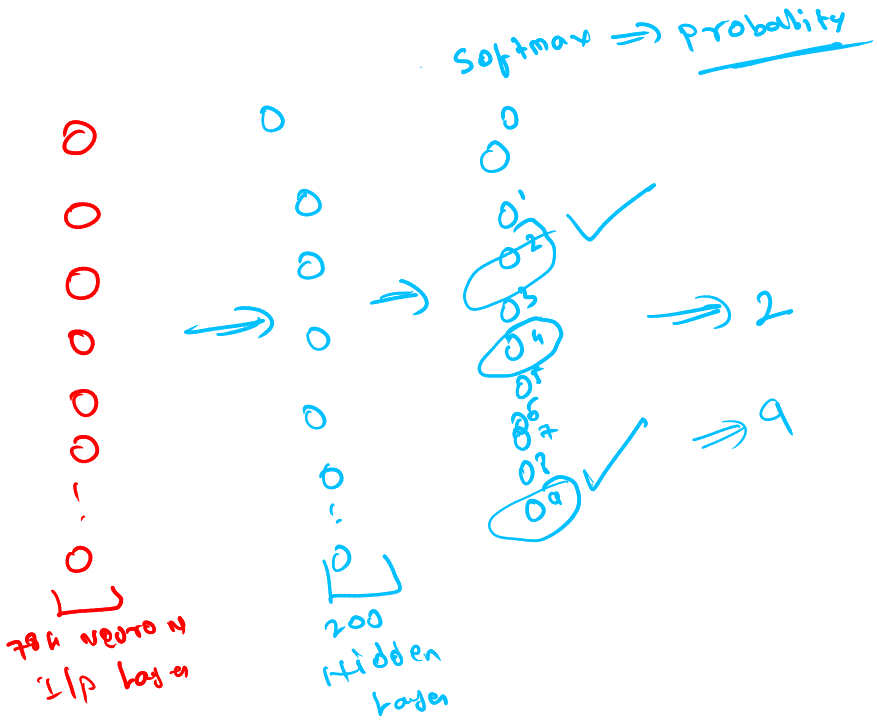




Targe — 2 classes — Binary

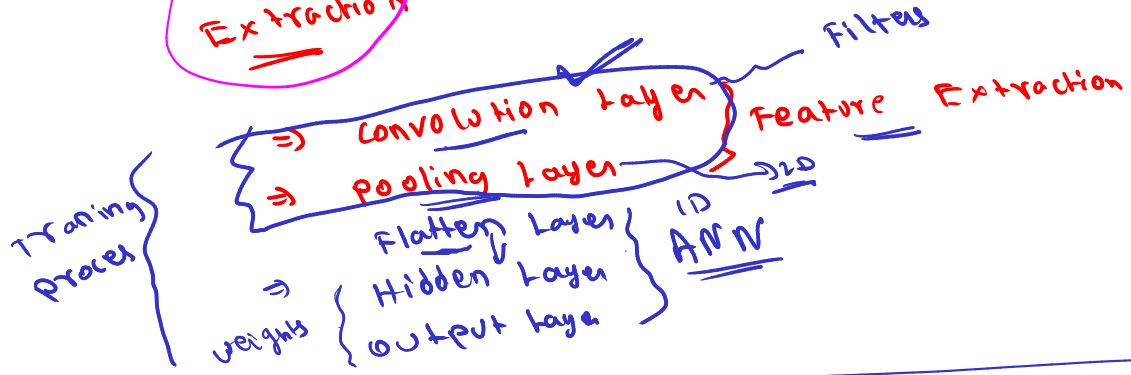
→ classes — Multiclass classification





CNN — 1997

Feature Extraction + ANN

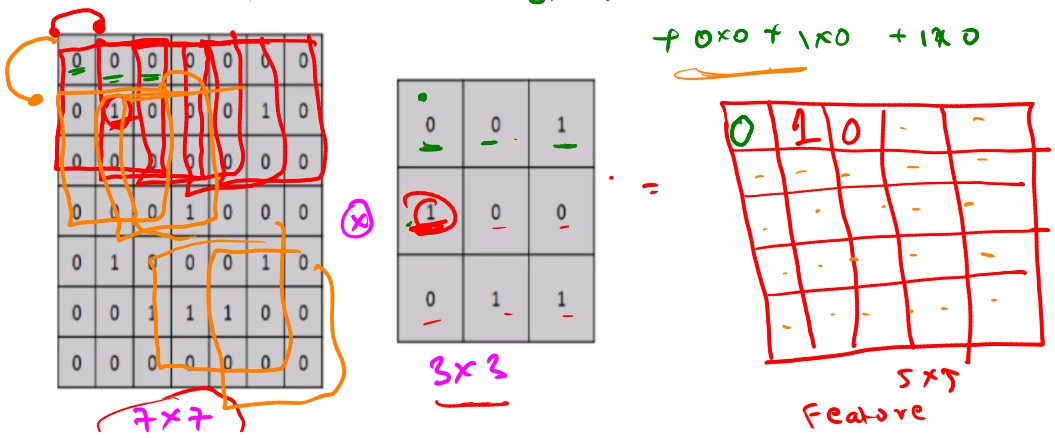


Filters → Extracting features

⇒ The model will learn the filters

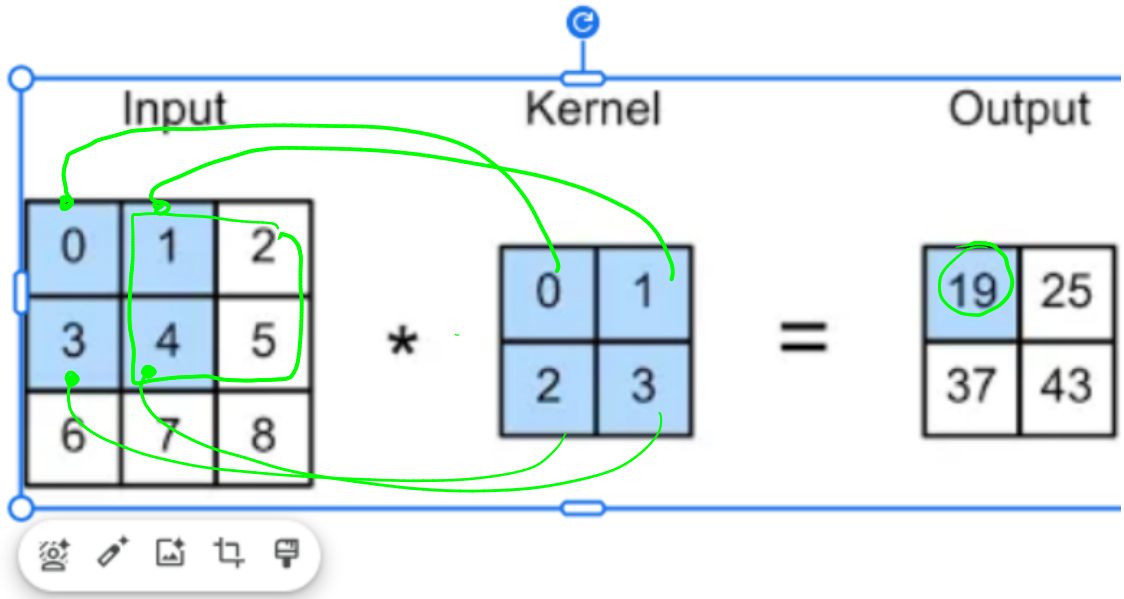
Convolution

Input Image → Filter

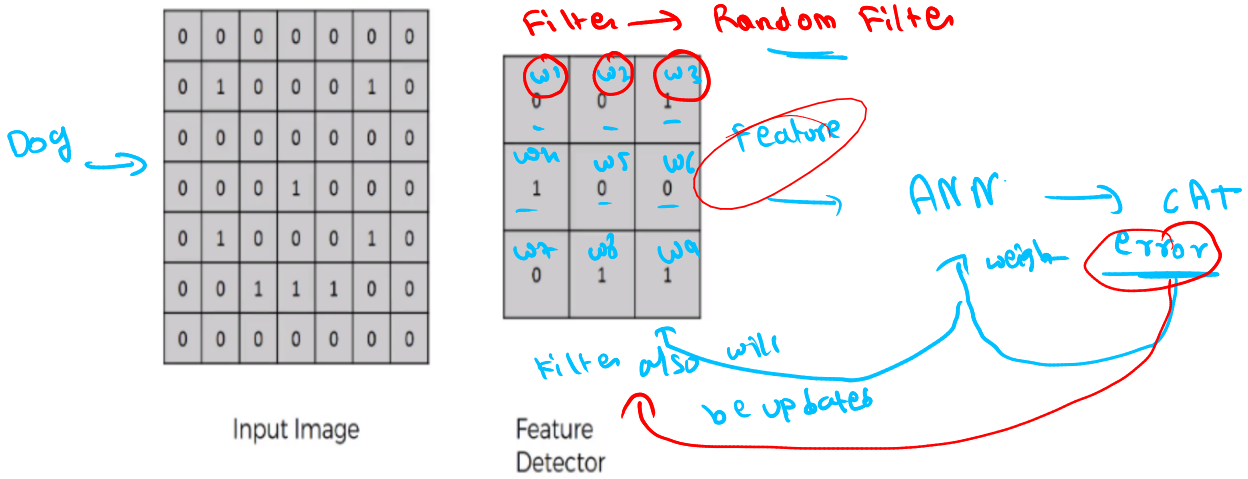


start with Random Filter

0×0 + 0×0 + 1×0 + 1×0 + 0×1 + 0×0 + 0×0 + 1×0 + 1×0



Filter / Feature Detector → extract the Important features from the Image



In CNNs During Back propagation → Filter will be updated

weights → error

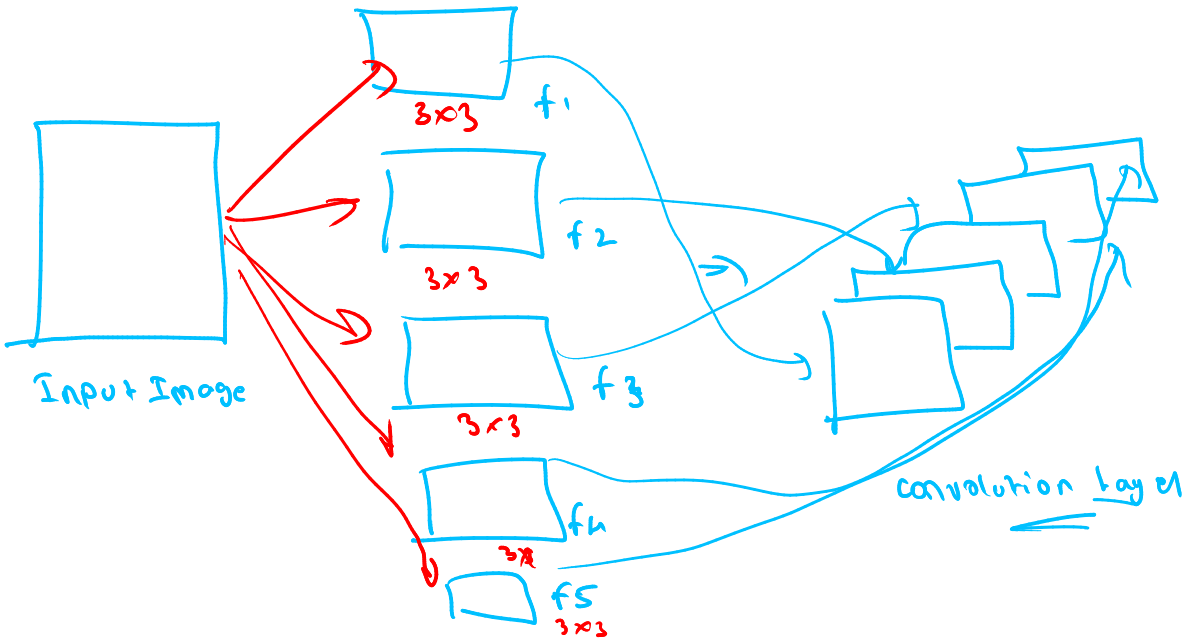
error – e

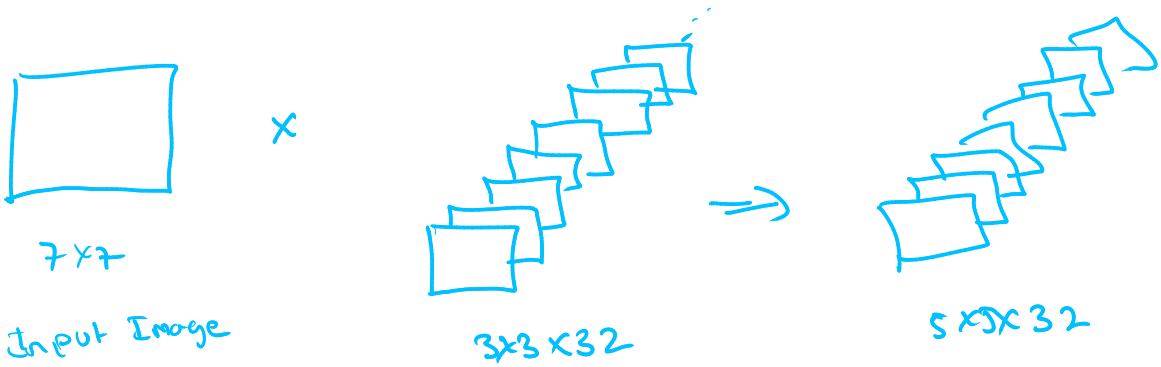
$$w = w - \alpha \cdot \frac{\partial e}{\partial w}$$

$$w_1 = w_1 - \alpha \cdot \frac{\partial e}{\partial w_1}$$

$\frac{\partial e}{\partial w}$ → How much is the error effected by the change in weight w_1

$\frac{\partial e}{\partial w}$ error ↓





For

Formulat to compute feature map size

Input Image = $n \times n$ $= 7 \times 7$

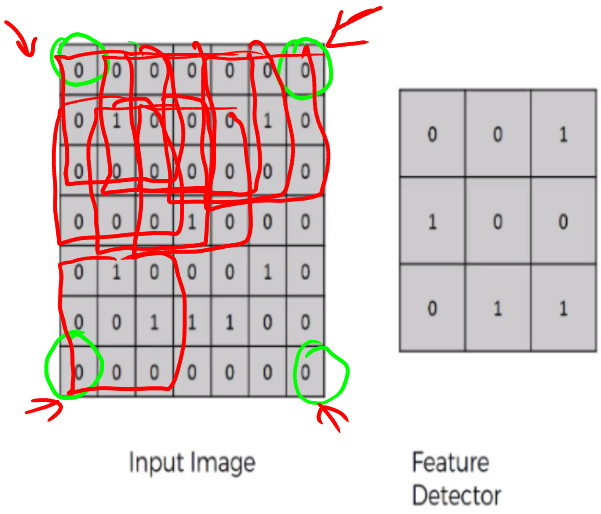
filter size = $f \times f$ $= 3 \times 3$

stride = $s \times s$ $= 1 \times 1$

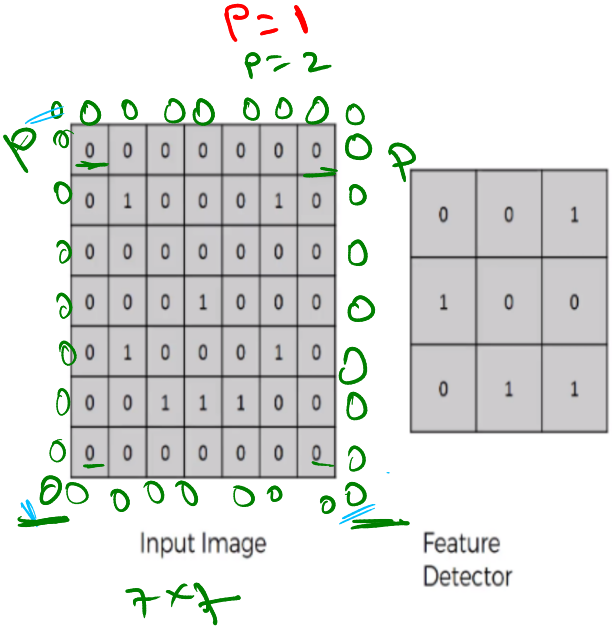
$$f' \times f' = \left(\frac{n-f}{s} + 1\right) * \left(\frac{n-f}{s} + 1\right) \quad (W \times H)$$

$$= \left(\frac{7-3}{1} + 1\right) * \left(\frac{7-3}{1} + 1\right) = 5 \times 5$$

⇒ Padding



1. Corner feature could be ignored
2. output size is shrinking



Input $n \times n = 7 \times 7$

filter $f \times f = 3 \times 3$

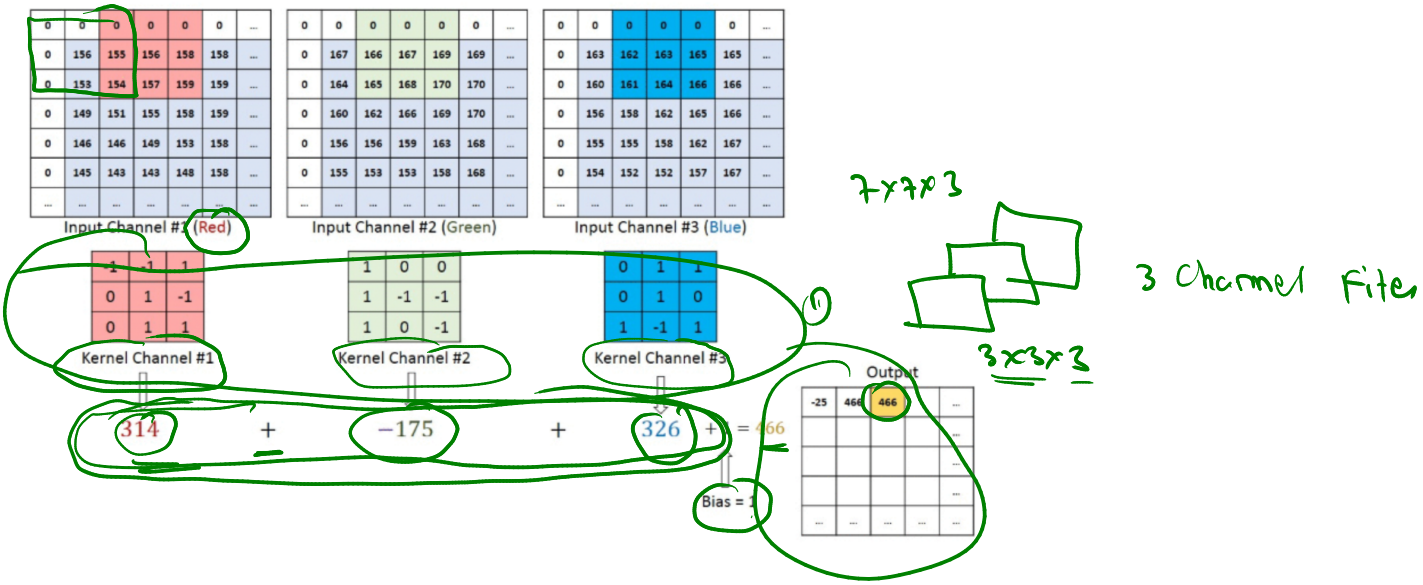
stride $s \times s = 1 \times 1$

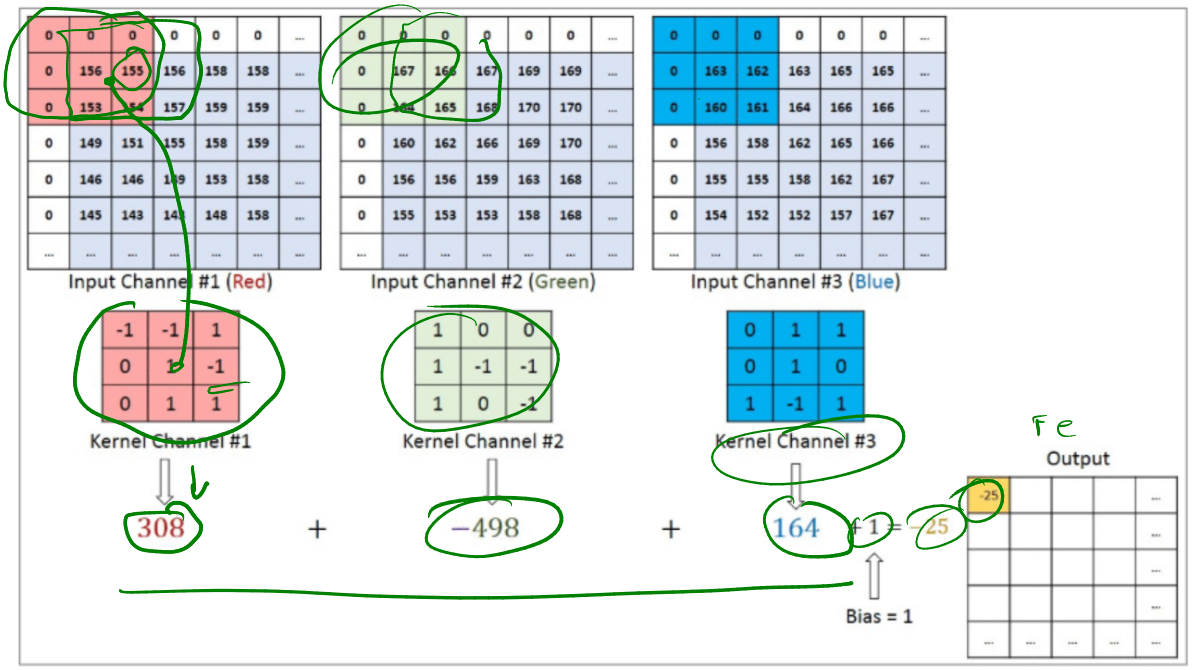
padding $p \times p = 1 \times 1$

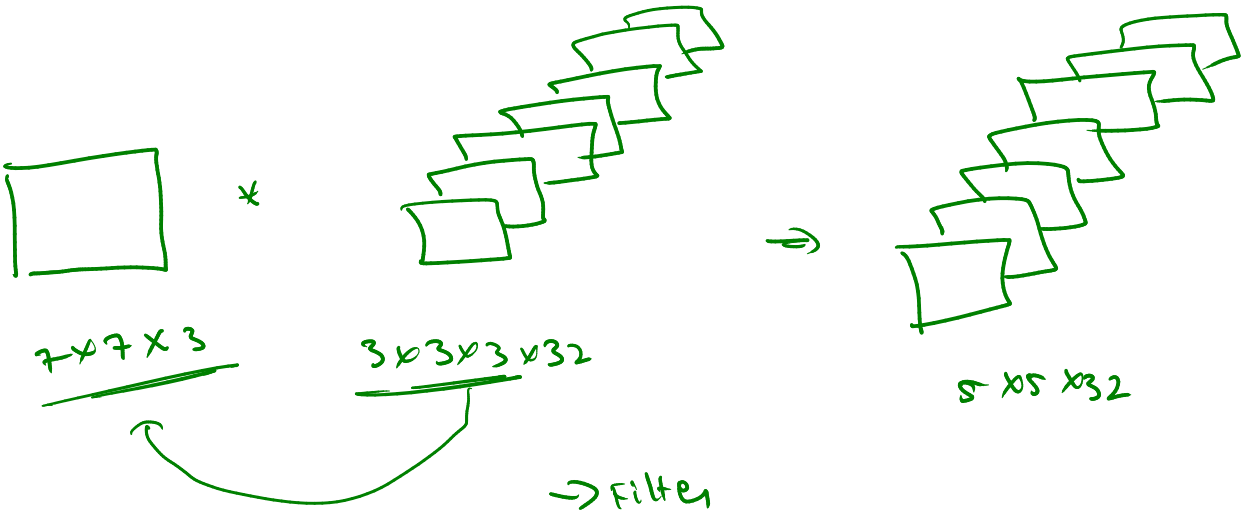
$$\frac{n + 2p - f}{s} + 1$$

$$= \frac{7 + 2 - 3}{1} + 1$$

$$= 7 \times 7$$







→ Filter
→ stride
→ padding
→ multiple Filter
→ conv in RGB Image

Pooling

overfitting / spacial Distortion



each pixel is a feature

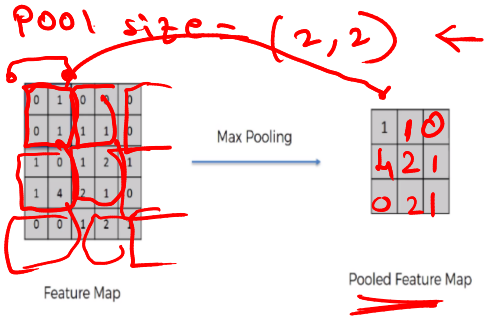
max pooling

mean/avg pooling

sum pooling

apply pooling feature maps

max Pooling



shape of feature map = nf = 5 x 5

pool size $p' = 2 \times 2$

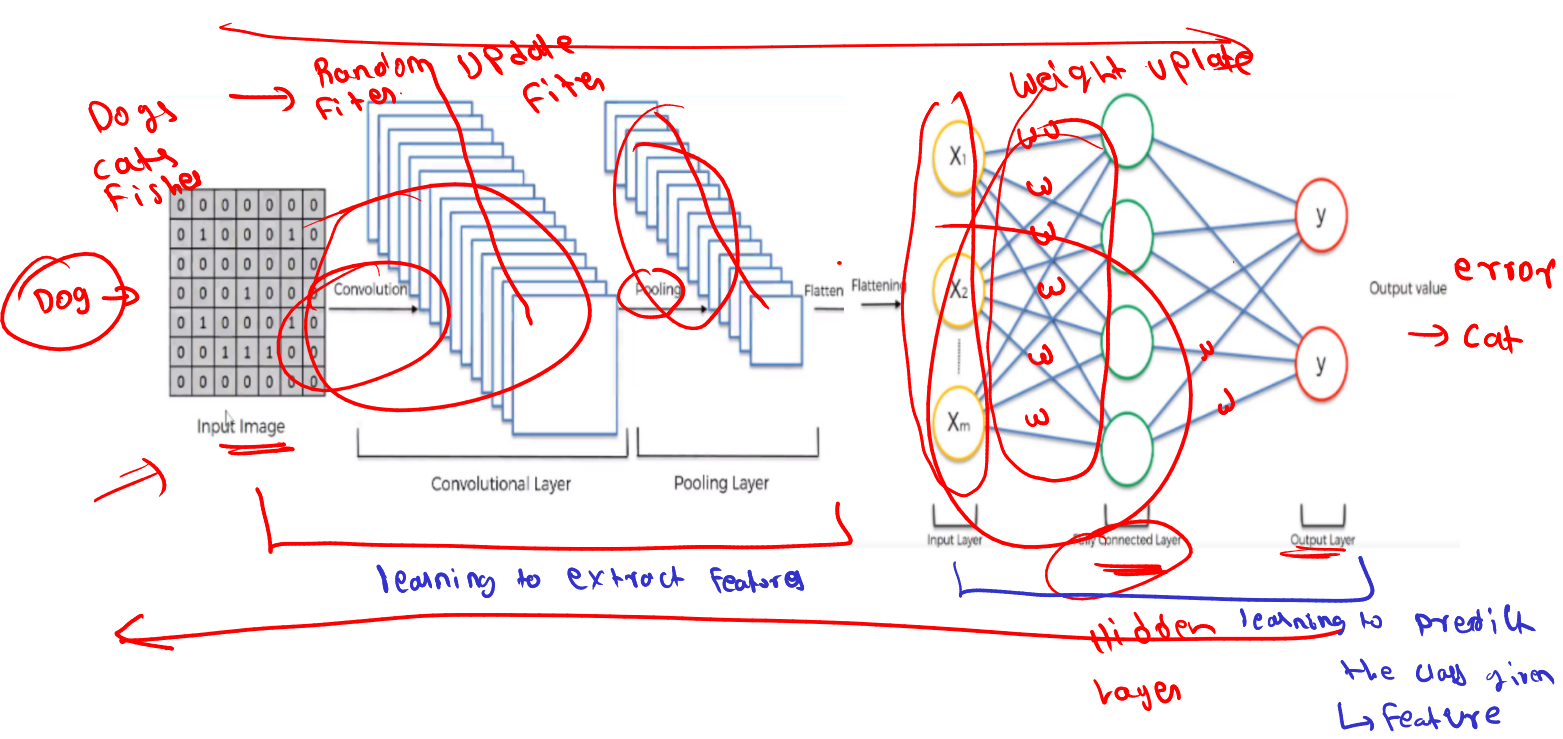
stride $s' = 2 \times 2$

$$\left(\frac{nf - p'}{s'} + \right) = \left(\frac{5-2}{2} + 1\right)$$

$$= 2.5 = 3$$

$$= (3 \times 3)$$

$$= 3 \times 3 \times 32$$


Dogs
cats
Fishes
Random Filter
Update Filter
Weight update
Dog
Input Image
Convolution
Pooling
Flatten
Flattening
Convolutional Layer
Pooling Layer
X1
X2
Xm
y
y
Output value
error
Cat
Input Layer
Fully Connected Layer
Output Layer
learning to extract features
Hidden layer
learning to predict the class given feature